

Trash Detection On Water Channels

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Abstract. Rivers and canals flowing through cities are often used illegally for dumping trash that contaminates freshwater channels, causes blockage in sewerage leading to urban flooding. The dumped trash is often found floating on the water surface. We propose to automatically identify this trash through visual inspection with the eventual goal of quantification, an early warning system to avoid blockages and urban flooding. The trash could be disfigured, partially submerged, or clumped together with other objects which obscure its shape and appearance. Thus, we consider surface trash as a blob detection problem that could either be solved as object detection or image segmentation or both. To this extent, we evaluate and compare several deep-learning-based object detection and segmentation algorithms. Unlike ocean trash, to the best of our knowledge, there is no large dataset on urban trash on water channels. Thus, using IoT-based camera nodes at multiple water channels, we collected a large dataset containing 48,450 trash objects annotated for both bounding box and segmentation (the dataset will be made publicly available¹). In addition, we also propose modifications in state-of-the-art detection and segmentation algorithms to cater to an issue such as partially submerged, varying object sizes, and edge-based computing.

Keywords: Trash · Object Detection · Segmentation.

1 INTRODUCTION

Every year millions of tons of trash, particularly plastic, are discarded globally that pollute our lands, rivers, and oceans. It causes environmental as well as economic repercussions. According to the United Nations world water development report [5], annually about 3.5 million people, mostly children, die from water-related infections. Almost all research on plastic contamination in water systems focuses on oceans. However, the source of this contamination, particularly in Asia and Africa, is the trash dumped into fresh and sewerage water channels, especially those passing through urban centers. These water channels transport this trash to rivers and finally to sea due to inadequate disposal and handling of landfill, industrial and general wastes [12].

Thus it is essential to identify the trash objects such as papers, card-boards, food residuals, plastic bottles, and bags, etc. which are present in freshwater ecosystems. In this work, we propose to identify these floating trash objects through a camera-based inspection system that will help in forming an index of water contamination by measuring the number of trash objects and water surface area covered by it. The identification

¹ Dataset is available at <https://cvlab.lums.edu.pk/watertrash/>



Fig. 1: Sample camera views from the collected dataset. The variation in views, shadows of overpass bridge, reflection of buildings and presence of vegetation is clear in these images.

and quantification of trash would then be used to remove trash via skimmer boats or by notifying concerned authorities for appropriate actions.

The existing work on vision-based approaches for the detection of trash could be divided into three categories i) Classification of trash in a controlled environment, applicable at waste recycling plants [22, 29] ii) Detection of piles of trash, usually illegally dumped in cities [16, 18] iii) Detection of sparse trash (street trash or marine litter) [7, 10, 11, 15]. In this paper, we introduce a fourth category of detecting visual trash floating on the water channels, especially drainage canals. Different from the above-discussed studies, our problem focuses on surface trash present in canals running through dense urban areas.

There are many solutions to perform automatic waste classification in controlled environments such as recycling plants [2, 22, 29]. For comparatively less managed scenarios such as streets in the urban center, convolution neural network based object detectors are used [18]. This work identifies street trash by performing object detection on images obtained from an RGB camera mounted on a dashboard of a vehicle [18].

Image segmentation has also been used to classify trash pixels. Kelo *et al.* [11] uses unmanned aerial vehicles (UAV) to map the trash on beaches. This work first computes a point cloud using photogrammetry and then applies CNN based segmentation strategy for trash localization. Similarly, SpotGarbage [16] performs segmentation of trash on publicly available images such as those obtained from Bing Maps or through crowdsourcing (TACO Litter dataset [17]). Hyper-spectral images [31] and RGB-D images [28] obtained via depth sensor have also been used to address the problem of trash identification. Similarly, deep detection networks are used for localization of marine litter [7, 10].

Most of the existing methods are either focused on street litter or ocean trash, recently a solution is proposed to monitor plastic contamination in rivers [25]. This work



Fig. 2: Sample cropped images from our dataset showing 10 challenging scenarios present in the fresh water and drainage canals.

first collects a dataset of 1272 images and then perform off the shelf image segmentation and classification for the identification of plastics. We instead study fresh and sewerage channels that are the source of contamination for rivers and oceans. Considering challenges; such as varying shape and appearance irregularities, trash localization can be considered as a *blob detection* problem and provide a comparative evaluation of various object detection and segmentation approaches. We also propose modifications in state-of-the-art detection and segmentation algorithms to cater for issues such as weak objects (such as partially submerged) and varying object sizes, we introduced log-attention, data imbalance is handled by replacing loss function with focal loss [30] and computational efficiency is improved through depth-wise separable convolution [9]. Furthermore, we present an IoT-based multi-camera setup deployed on these water channels for continuous data collection and monitoring. As a result, we also provide a comparatively large dataset containing 13,500 images and 48,450 object annotations (bounding boxes and segmentation mask).

The remainder of this paper is organized as follows. Section 2 introduces the dataset, comparative evaluation of object detection and segmentation approaches is discussed in Section 3 and Section 4 respectively. Finally, the paper is concluded in Section 5.

2 Dataset

2.1 Collection and annotation

To obtain an extensive dataset containing wide-variety of realistic scenarios, we selected multiple sites across the city and performed video recording during different day times, weather conditions, and at different types of localities (market, dense urban, peri-urban, etc). After removing noisy samples, a total of 1,35,00 images were extracted from these videos. These images were annotated for bounding boxes using LabelImg [24] and for

Table 1: Distribution of small, medium and large instances (trash objects) in annotated images as per COCO [4] standard.

Size	No. of Objects	Area (pixels)
Small	11,214	$\text{area} \leq 32^2$
Medium	32,078	$32^2 < \text{area} \leq 96^2$
Large	5,158	$\text{area} > 96^2$
Total	48,450	$\text{area} > 7^2$

segmentation using LabelMe [21]. Plastic bottles, plastic bags, papers, cardboards, and food residuals, etc. floating on the water surface either fully or partially visible (submerged) were regarded as garbage. We intentionally do not annotate micro-particles, plant leaves, and air bubbles. Thus, 4, 88, 98 instances were opted as trash objects. We consider 1, 25, 00 samples as train-validation set and the remaining 1, 000 images are used as a test set. Apart from images from the same sites, the test set also contains images that are from a different distribution i.e different view angles and sites. The example images are shown in Fig. 1, the dataset contains several challenging scenarios that are discussed next.

2.2 Challenges in dataset

No fixed geometry: Objects in the trash do not have a defined geometrical shape and the shapes also change over time. For example, floating plastic bags can have any shape which also changes with time. The trash also contains objects which are damaged either before dumping or in the process such as plastic bottles or juice packets are collapsed and punctured and have wrinkled labels (see Fig. 2(a)).

Partial visibility of objects: Many of the objects are partially submerged in water and it has also been observed that due to the flow of water, objects sometimes sink and then resurface after a while. It has been observed that the texture of objects such as cardboards (see Fig. 2(b)), make them very difficult to be detected.

Reflections: Since the dataset was collected in dense urban areas, the reflections of both static and moving objects cause a significant change in the appearance of the water channel. Static objects include buildings and electricity poles, which depending upon the camera view-point and time of the day, may cover a significant portion of the water surface. Moving objects such as flying birds reflect over the water surface, which often exhibits very similar features to that of flowing garbage. Three samples of reflection are demonstrated in Fig. 2 (c), (d) and (f).

Shadows: Similar to reflections, the shadows of buildings and poles, especially during morning and afternoon hours, create unfavorable conditions for deep detectors.

Table 2: Comparison with existing trash datasets. (Key: annot.: annotation, BBox: Bounding box for object detection, Mask: segmentation mask).

Dataset	Type	# images	# classes	# instances	annot. type
TrashNet [29]	Indoor	2500	6	2500	Classification
VN-trash [26]	Indoor	5904	3	5904	Classification
Street [18]	Road	469	6	1421	BBox
TACO [17]	Multi	1500	60	4784	Mask
MJU-Waste [28]	Multi	2475	1	2524	Mask
Trash-det [7]	Marine	5720	3	-	BBox
TrashCan [10]	Marine	7212	22	12480	Mask
River Plastic [25]	Rivers	1272	1	14892	Mask
Our	Water Channel	13500	1	48450	BBox, Mask

Water opacity: The color of the water surface varies with the ratio of chemical discharge from industries that alters the opacity. It not only changes the texture of the water surface but also affects the reflection and refraction of sunlight.

Wide density of objects: As shown in Fig. 2(g)-(i), there are sparse objects as well as dense piles of trash. For sparse objects, each particle is considered as a different object, whereas the piles of trash are considered as a single object. Thus, a wide range of variable sizes of objects has been considered. But, the micro-particles as given in Fig. 2 (g), are ignored since they are smaller in size and when the image is resized to less than 640 pixel, they become invisible.

Air bubbles: Sometimes sewer gases are produced through the decomposition of organic household products or industrial waste, which results in air bubbles forming on the water surface as depicted in Fig. 2 (j). The bubbles look similar to sub-merged objects or damaged cardboards and result in false detection from the networks.

2.3 Comparison with existing datasets

Table 2 compares existing datasets pertaining to trash detection in different scenarios. TrashNet [29] and VN-trash [26] contain images taken in controlled environments, therefore objects are clearly visible and each image contains only one object. TACO [17] is a cloud-sourced trash RGB images, MJU-Waste [28] contains point-cloud of trash objects, and Street [18] is collected through car dashboard camera. These datasets contain fewer image samples, therefore insufficient for generalization of deep learning models. Trash-det [7] and TrashCan [10] are underwater marine debris datasets. The recent River Plastic [25] dataset also contains very few images of specific locations and fixed lighting conditions. To the best of our knowledge, we have yet to find a large dataset that entails the study of trash in water channels. Our dataset is significantly larger than the existing datasets.

Table 3: Comparative evaluation of the state-of-the-art object detection techniques. (Key: AP: Average Precision, AP^S , AP^M and AP^L : AP small, medium and large size objects, M: Millions).

Model	#Param	AP^S	AP^M	AP^L	mAP
YOLO-v3 [19]	61.5M	2.7	11.4	33.7	35.1
YOLO-v3-Tiny	8.6M	0.4	2.8	14.5	9.0
RetinaNet [14]	55.4M	4.6	23.0	42.6	43.4
PeleeNet [27]	4.8M	3.5	10.9	30.1	29.9
M2Det (VGG) [32]	132M	5.2	19.8	35.2	45.8

3 Trash Object Identification

Most of the well-known object detectors [14, 19, 27, 32] are designed for general applications, especially for urban scenarios such as those related to surveillance and self-driving cars. Although they perform better on relevant benchmark datasets such as MS-COCO [4] and Pascal-VOC [6], however, the difficulty of trash detection is significantly higher due to the challenges present in the dataset.

We retrained five different detectors on our dataset namely YOLO-v3 [19], YOLO-v3-Tiny, RetinaNet [14], PeleeNet [27] and M2Det (VGG) [32] and demonstrate their results in Table 3. Overall, M2Det with VGG-16 as backbone outperforms all the other models whereas YOLO-v3-Tiny fails to learn complex information. The better performance of M2Det is due to its multi-level feature pyramid network (MLFPN). RetinaNet and PeleeNet have moderate performance. PeleeNet despite being a smaller network produces results comparable to RetinaNet and YOLO-v3. It can also be validated from Fig. 3 (c)-(f), where both scenes contain different water textures and have a reflection from a pole and buildings, most of the objects are detected.

3.1 Analysis on object sizes

In order to find out the performance dependency on object sizes, we evaluated the trained networks at three scales of objects provided in Table 1. Table 4 demonstrates detection of smaller objects is difficult than medium and large objects. This is probably due to prior training of models for object detection task on the Pascal VOC dataset. Therefore, the information of objectness was retained for large objects whereas learned information about small objects is forgotten and learned again. As claimed by M2Det [32], it has outperformed on small as well as large objects. RetinaNet is second best as it is built upon a feature pyramid network (FPN). Although PeleeNet is a smaller model, yet its performance is comparable to YOLO-v3, even it has a better AP score for small objects. It would also be a better choice for deployment on an embedded computing platform.

3.2 Handling smaller and partially visible objects

To address the issue of poor performance on sub-merged, partially visible, and smaller objects by detection models, an attention layer is employed that tries to focus on weaker

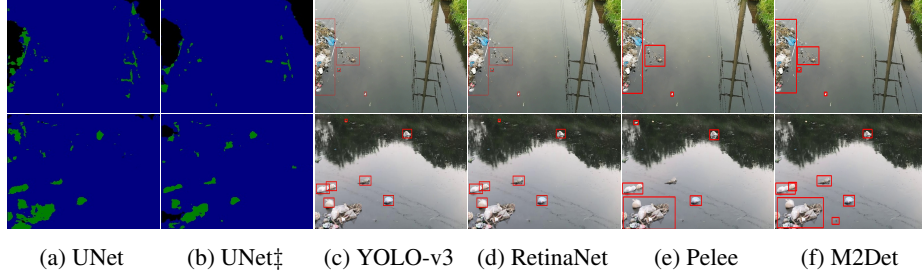


Fig. 3: Qualitative results of trained model: (a), (b) are segmentation output from vanilla UNet and UNet with separable convolution layer, respectively where green represents trash pixels, black and blue are land nad water pixels. (c)-(f) show the trash detected by with object detectors.

activation. Attention based on *Sigmoid* and *Softmax* activation functions perform well for segmenting the pixels near boundaries of the objects but do not work in case of object detection. So, we amplify weaker activations through *Log* attention that can be defined as:

$$f_{i+1} = f_i \times \log(\text{ReLU}(f_i) + 1), \quad (1)$$

where f_i is output of i^{th} layer, and $i = 0, \dots, N$ and N is number of layers. Here, ReLU discards the negative values in the activations and bias 1 shifts it one scale up, making it possible to compute $\log$. The derivative of this attention layer would be:

$$\nabla f_{i+1} = \begin{cases} \log(f_i + 1) + \frac{1}{f_i + 1} & f_i \geq 0 \\ 1 & otherwise \end{cases} \quad (2)$$

Table 4 demonstrates improvement in performance due to introduced log attention layer. YOLO-v3 has been improved for all objects, however, PeeleNet does not improve because it is a smaller model and has limited learning capacity.

4 Trash Pixel identification

Trash may have blob-like irregular shape and appearance and it also breaks into smaller pieces as it floats through water channels. Instead of object-level analysis, pixel-level analysis via image segmentation results in better localization in such cases. Thus in this

Table 4: Effect of Log Attention Layer on results of object detection. (Key: mAP: Mean Average Precision).

Model	AP^S	AP^M	AP^L	mAP
YOLO-v3+Attn	3.5	11.6	34.5	35.3
PeeleNet+Attn	3.4	10.3	31.5	29.4

work, we also provided an analysis of popular semantic segmentation models on our dataset. We first trained vanilla models (UNet [20] and SegNet [1]) that use standard convolution. From Table 5, it is clear that standard segmentation models show better performance as each pixel of trash is classified. However, these networks do have a larger number of parameters and are thus not suitable for continuous monitoring.

4.1 Reducing model size for real time applications

We propose a modified UNet and SegNet architectures and improve their inference time by introducing depth-wise separable convolution [9], which can reduce the parameters in standard convolution [3, 8, 13]. As depicted in Table 5, when we replace standard convolution with separable convolution in alternate layers (Sep[†]), parameters of both models reduce to approximately half of the original and when all standard convolution layers are replaced with separable convolution layers (Sep[‡]), the models shrink by almost 10 times resulting 2x improvement in terms of frames per second. From Fig. 3, it is apparent that the segmentation map of before and after reducing the size of the UNet model are quite similar.

4.2 Addressing Class Imbalance

In segmentation, class distribution in each image instance varies. Sometimes one of the class has an extremely low number of pixels in the whole dataset, as in our case where the trash samples are almost 1% and water is more than 89%. To cater to this and to put specific emphasis on relatively difficult examples, we further modified the UNet and SegNet models and introduced focal loss [30] defined as:

$$CB_f(p, y) = \alpha \sum_{j=1}^C (1 - p_i) \log(p_i), \quad (3)$$

where, p is prediction of model, y is ground truth label, C is number of classes and α is weighting factor. To address the problem of missing data, the inverse probability

Table 5: Performance of standard segmentation models along with their smaller versions achieved by employing separable convolution layers. Sep[†] is model with half of its layers employ separable convolution and Sep[‡] has all layers as separable convolution. FPS = Frame per second, M = millions.

Measure	UNet			SegNet		
	Vanilla	Sep [†]	Sep [‡]	Vanilla	Sep [†]	Sep [‡]
#Param (M)	31	18	3.9	29	16.8	3.3
FPS	8	12	14	8	12	16
Acc (Trash)	87%	72%	57%	88%	70%	69%

Table 6: Addressing class imbalance on UNet vanilla and its variants. CE is cross entropy loss, CB_f and CB_{ipw} are class balanced focal loss and our inverse probability loss functions.

Model	CE	CB_f	CB_{ipw}
UNet vanilla	87%	88%	92%
UNet Sep†	72%	78%	82%
UNet Sep‡	57%	75%	81%

weighting (ipw) [23] is used which removes the bias by penalizing the estimator. It can be defined as:

$$CB_{ipw}(p, y) = \frac{\sum_{i=1}^C n_i}{n_c} \sum_{j=1}^C (1 - p_i) \log(p_i), \quad (4)$$

where, n_c number of pixels of c^{th} classes in whole dataset. In our case, trash is seldom in the image instances, inverse probability weighting would be better class weighting scheme.

These loss functions as depicted in Table 6 have significantly improved the accuracy of all models overcoming a drop in accuracy due to separable convolution. When UNet Sep‡ is trained using CB_{ipw} loss, despite the model is 800% times smaller there is only 10% drop in accuracy, the model applicable for real-time applications.

5 Conclusions and Future work

This paper presents a new category of visual trash detection and provides a large benchmark trash dataset. A dataset of trash floating on canal surface in dense urban areas is collected and annotated. Due to various challenges in the dataset, both object detection and segmentation approaches were analyzed. Five recent and popular object detection models and six variants of segmentation models were trained. This paper studies the difficulty in detection of submerged, partially visible, and smaller objects and propose improvements through $\log$ attention, reduce model complexity through separable convolution layer for real-time deployment of models, and addresses the class imbalance in segmentation through focal loss functions. Overall, the detection of floating trash especially in water channels in urban areas is a challenging task and an emerging area of research. The dataset and comparative analysis provided in this work will serve as a stepping stone towards finding a solution to this problem. In the future, we plan to merge both object detection and segmentation approaches through architectural modifications such that it would force the model to learn better features of weak objects. We also plan to increase the dataset size and categorize trash into different classes such as organic, plastic, paper, leaves, metals, etc.

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